

**Course Code : ECT 359-3**

**MPNO/MS – 24 / 1669**

**Sixth Semester B. Tech. (Electronics and Communication  
Engineering) Examination**

**BIOMEDICAL ELECTRONICS**

Time : 3 Hours ]

[ Max. Marks : 60

**Instructions to Candidates :—**

- (1) All questions are compulsory.
  - (2) All questions carry marks as indicated against them.
  - (3) Assume suitable data and illustrate answers with neat sketches wherever necessary.
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1.     (a) State and explain the factors to be considered while designing a medical instrumentation system. 6(CO1)  
       (b) With suitable schematic, explain the application of piezoelectric transducer in medical electronics. 6(CO1,2)
  
  2.     (a) Mention the origin and features of the signal generated by heart. Explain the working of ECG recorder with emphasis to various lead configurations. 6(CO2,4)  
       (b) What is pacemaker ? List pacemaker types and compare them. 6(CO2)
  
  3.     (a) Discuss the operation of dc defibrillator with suitable schematic diagram and mention its use. 6(CO2)  
       (b) How EEG test is performed ? Support your answer with relevant schematic diagram. 6(CO2,4)
  
  4.     (a) With the neat diagram, explain the principle of CT scan imaging system. 6(CO3)

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**Contd.**

- (b) Discuss the medical equipments and their need in intensive care unit (ICU). 6(CO3,4)
5. (a) Explain the physiological effects of electrical shock to human body. How patient electrical safety is taken care in a hospital ? 6(CO5)
- (b) Which machine is crucial during open heart surgery ? Write the operational details of heart-lung machine. 6(CO5)

